

LEC29 - Alternating Series

Warm Up Problem

Which of the following are alternating series?

1. $\sum_{n=1}^{\infty} \frac{(-1)^n}{\tan(n)}$ $\Rightarrow$ Terms do not always alternate
2. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ $\Rightarrow$ Terms do not alternate
3. $\sum_{n=1}^{\infty} \cos(n)$ $\Rightarrow$ Terms do not alternate
4. $\sum_{n=1}^{\infty} \frac{\cos(\pi n)}{n^2}$ $= \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$, which alternates

Alternating series test

If the alternating series $\sum_{n=1}^{\infty} (-1)^n b_n$ or $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ satisfies these two properties

- i) $0 \leq b_{n+1} \leq b_n$ for all $n \geq 1$ (Non-negative and decreasing)
- ii) $\lim_{n \rightarrow \infty} b_n = 0$

then the series converges

Example

Does $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(n-1)!}$ converge or diverge?

$$\Rightarrow 0 \leq \frac{1}{n!} \leq \frac{1}{(n-1)!}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{(n-1)!} = 0$$

$\Rightarrow$ Since the series is an alternating series, by the AST it converges

Alternating Series Remainders

Suppose the alternating $\sum_{n=1}^{\infty} (-1)^n b_n$ converges by the AST

- Let S be the sum of the series
- Let S_N be the N^{th} partial sum of the series $S_N = \sum_{n=1}^N (-1)^n b_n$
- Let $R_N = S - S_N$, the remainder or error between S and S_N

$$\text{Then } |R_N| \leq b_{N+1}$$

Example

Find a rational approximation of S accurate to within 0.001. How many terms are needed?

$$b_1 = \frac{1}{0!} = 1 \quad b_2 = \frac{1}{1!} = 1 \quad b_3 = \frac{1}{2!} = \frac{1}{2}$$

$$b_4 = \frac{1}{3!} = \frac{1}{6} \quad b_5 = \frac{1}{4!} = \frac{1}{24} \quad b_6 = \frac{1}{5!} = \frac{1}{120}$$

$$b_7 = \frac{1}{6!} = \frac{1}{720} \quad b_8 = \frac{1}{7!} = \frac{1}{5040} \Rightarrow |0.001| \leq b_8 \Rightarrow 8 = n+1 \Rightarrow n = 7$$

Absolute Convergence

A series $\sum_{n=1}^{\infty} a_n$ is called absolutely convergent if

the series $\sum_{n=1}^{\infty} |a_n|$ converges

Example

Is $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ absolutely convergent?

$$\Rightarrow a_n = \frac{(-1)^n}{n^2}$$

$$\Rightarrow |a_n| = \frac{1}{n^2}, \text{ where } \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges}$$

Theorem 5.15

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges

So any series which is absolutely convergent is also conditionally convergent

LEC30 - Alternating Series Continued

Warm Up Problem

Which of these series converge?

1. Geometric series with $r = \frac{1}{2}$
2. Harmonic series
3. Alternating Harmonic series

Absolute and Conditional Convergence

- $\sum_{n=1}^{\infty} a_n$ is called absolutely convergent if the series $\sum_{n=1}^{\infty} |a_n|$ converges
- A series which is convergent but not absolutely convergent is called conditionally convergent
- If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ must converge

Divergent	Conditionally Convergent	Absolutely Convergent
$\sum a_n$ D	$\sum a_n$ C	$\sum a_n$ C
$\sum a_n $ D	$\sum a_n $ D	$\sum a_n $ C

Example

Does $\sum_{n=1}^{\infty} (-1)^n \frac{e^n}{3^n}$ converge absolutely, converge

conditionally, or diverge?

$$\Rightarrow \lim_{n \rightarrow \infty} \left[\frac{e^n}{3^n} \right] = \infty$$

$\Rightarrow$ Therefore, by the divergence test, the series diverges

Example

Does $\sum_{n=1}^{\infty} (-1)^n \frac{5n^2 + 7}{2n + 3n^4}$ converge absolutely, converge

conditionally, or diverge?

$$\Rightarrow \text{Use limit comparison test with } a_n = \frac{5n^2 + 7}{2n + 3n^4}, b_n = \frac{5n^2}{3n^4} = \frac{5}{3n^2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left[\frac{5n^2 + 7}{2n + 3n^4} \cdot \frac{3n^2}{5} \right] = 1, \text{ so } a_n \text{ and } b_n \text{ either both converge or both diverge}$$

$$\Rightarrow \text{Since } \sum \frac{5}{3n^2} \text{ converges, } a_n \text{ must converge}$$

$\Rightarrow$ Therefore, the series is absolutely convergent

$$\text{So } \sum_{n=1}^{\infty} (-1)^n \frac{5n^2 + 7}{2n + 3n^4} \text{ must also converge}$$

Example

Which of the following statements are true?

1. If $a_n > 0$ for all n , $\sum_{n=1}^{\infty} a_n$ cannot be conditionally convergent
False Consider $a_n = \frac{1}{n} > 0$ for all n , but a_n is conditionally convergent
2. If $a_n > 0$ and $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} (-1)^n a_n$ converges too
True by the definition of absolute convergence
3. If $\sum_{n=1}^{\infty} a_n$ is an alternating series, then it converges
False Consider $a_n = (-1)^n \cdot 2^n$, which diverges since $\lim_{n \rightarrow \infty} 2^n \neq 0$